

CSE370 : Database Systems

Assignment 03 | Spring 2025

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No 1 Query (as Plain Text)	SELECT customer.customer_name, borrower.loan_number FROM customer INNER JOIN borrower ON customer.customer_id = borrower.customer_id INNER JOIN loan ON borrower.loan_number = loan.loan_number WHERE loan.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [Bank_22201344]> SELECT customer.customer_name, borrower.loan_number -> FROM customer -> INNER JOIN borrower ON customer.customer_id = borrower.customer_id -> INNER JOIN loan ON borrower.loan_number = loan.loan_number -> WHERE loan.branch_name = 'Downtown'; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.002 sec)</pre>
No 2 Query (as Plain Text)	SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City FROM customer c1 INNER JOIN customer c2 ON c1.customer_city = c2.customer_city

	WHERE c1.customer_id < c2.customer_id;
No 2 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_22201344]> SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City -> FROM customer c1 -> INNER JOIN customer c2 ON c1.customer_city = c2.customer_city -> WHERE c1.customer_id < c2.customer_id; +-----+-----+-----+ Customer1 Customer2 City +-----+-----+-----+ Jones Hayes Harrison Smith Curry Rye Lindsay Adams Pittsfield Turner Green Stamford +-----+-----+-----+ 4 rows in set (0.001 sec) </pre>
No 3 Query (as Plain Text)	<pre> SELECT account.branch_name, SUM(account.balance * 0.04) AS total_interest FROM account GROUP BY account.branch_name; </pre>
No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_22201344]> SELECT account.branch_name, SUM(account.balance * 0.04) AS total_interest -> FROM account -> GROUP BY account.branch_name; +-----+-----+ branch_name total_interest +-----+-----+ Brighton 66.00 Downtown 20.00 Mianus 28.00 Perryridge 16.00 Redwood 28.00 Round Hill 14.00 +-----+-----+ 6 rows in set (0.001 sec) </pre>

No 4 Query (as Plain Text)	<pre>SELECT account.account_number, customer.customer_city, account.balance FROM account INNER JOIN depositor ON account.account_number = depositor.account_number INNER JOIN customer ON depositor.customer_id = customer.customer_id WHERE (customer.customer_city, account.balance) IN (SELECT customer.customer_city, MAX(account.balance) FROM account INNER JOIN depositor ON account.account_number = depositor.account_number INNER JOIN customer ON depositor.customer_id = customer.customer_id GROUP BY customer.customer_city);</pre>

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201344]> SELECT account.account_number, customer.customer_city, account.balance
-> FROM account
-> INNER JOIN depositor ON account.account_number = depositor.account_number
-> INNER JOIN customer ON depositor.customer_id = customer.customer_id
-> WHERE (customer.customer_city, account.balance) IN
-> (SELECT customer.customer_city, MAX(account.balance)
-> FROM account
-> INNER JOIN depositor ON account.account_number = depositor.account_number
-> INNER JOIN customer ON depositor.customer_id = customer.customer_id
-> GROUP BY customer.customer_city);
```

account_number	customer_city	balance
A-217	Harrison	750
A-215	Rye	700
A-222	Pittsfield	700
A-305	Stamford	350
A-201	Palo Alto	900

```
5 rows in set (0.001 sec)
```

No 5 Query
(as Plain Text)

```
SELECT loan.loan_number, loan.amount, customer.customer_name
FROM loan
INNER JOIN borrower ON loan.loan_number = borrower.loan_number
INNER JOIN customer ON borrower.customer_id = customer.customer_id
ORDER BY loan.amount DESC, loan.loan_number DESC
LIMIT 5;
```

No 5 SS (of Query & Output in Shell)	<pre>MariaDB [Bank_22201344]> SELECT loan.loan_number, loan.amount, customer.customer_name -> FROM loan -> INNER JOIN borrower ON loan.loan_number = borrower.loan_number -> INNER JOIN customer ON borrower.customer_id = customer.customer_id -> ORDER BY loan.amount DESC, loan.loan_number DESC -> LIMIT 5;</pre> <table><tr><th>loan_number</th><th>amount</th><th>customer_name</th></tr><tr><td>L-23</td><td>2000</td><td>Smith</td></tr><tr><td>L-15</td><td>1500</td><td>Hayes</td></tr><tr><td>L-14</td><td>1500</td><td>Johnson</td></tr><tr><td>L-16</td><td>1300</td><td>Adams</td></tr><tr><td>L-17</td><td>1000</td><td>Jones</td></tr></table> <pre>5 rows in set (0.015 sec)</pre>	loan_number	amount	customer_name	L-23	2000	Smith	L-15	1500	Hayes	L-14	1500	Johnson	L-16	1300	Adams	L-17	1000	Jones
loan_number	amount	customer_name																	
L-23	2000	Smith																	
L-15	1500	Hayes																	
L-14	1500	Johnson																	
L-16	1300	Adams																	
L-17	1000	Jones																	
No 6 Query (as Plain Text)	<pre>SELECT DISTINCT customer.customer_name FROM customer INNER JOIN depositor ON customer.customer_id = depositor.customer_id INNER JOIN account ON depositor.account_number = account.account_number INNER JOIN borrower ON customer.customer_id = borrower.customer_id INNER JOIN loan ON borrower.loan_number = loan.loan_number WHERE account.branch_name = 'Perryridge' AND loan.branch_name = 'Perryridge';</pre>																		

No 6 SS (of Query & Output in Shell)	<pre>MariaDB [Bank_22201344]> SELECT DISTINCT customer.customer_name -> FROM customer -> INNER JOIN depositor ON customer.customer_id = depositor.customer_id -> INNER JOIN account ON depositor.account_number = account.account_number -> INNER JOIN borrower ON customer.customer_id = borrower.customer_id -> INNER JOIN loan ON borrower.loan_number = loan.loan_number -> WHERE account.branch_name = 'Perryridge' AND loan.branch_name = 'Perryridge'; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.001 sec)</pre>
No 7 Query (as Plain Text)	<pre>SELECT customer.customer_name, SUM(loan.amount) AS total_loan FROM customer INNER JOIN borrower ON customer.customer_id = borrower.customer_id INNER JOIN loan ON borrower.loan_number = loan.loan_number GROUP BY customer.customer_id HAVING COUNT(borrower.loan_number) >= 2;</pre>

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201344]> SELECT customer.customer_name, SUM(loan.amount) AS total_loan  
-> FROM customer  
-> INNER JOIN borrower ON customer.customer_id = borrower.customer_id  
-> INNER JOIN loan ON borrower.loan_number = loan.loan_number  
-> GROUP BY customer.customer_id  
-> HAVING COUNT(borrower.loan_number) >= 2;
```

```
+-----+-----+  
| customer_name | total_loan |  
+-----+-----+  
| Smith        |         2900 |  
+-----+-----+  
1 row in set (0.001 sec)
```